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REVIEW

Multi-material glass additive manufacturing: a review of materials, techniques, and integration challenges

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Abstract

Glass offers unique advantages such as high optical transparency, thermal stability, and chemical resistance, making it indispensable in optics, microelectronics, sensing, and biomedical applications. However, its high melting point, brittle nature, and incompatibility with traditional manufacturing methods present significant fabrication challenges. Additive manufacturing (AM) introduces new possibilities for the glass industry by enabling the fabrication of customised, complex 3D structures and the integration of glass with other compatible materials. In particular, multi-material glass additive manufacturing (MMGAM) has emerged as a promising frontier, enabling both the spatial tuning of glass composition, dopants, and functionalities, as well as the co-fabrication of glass with dissimilar materials, such as polymers, metals, or ceramics, within a single component using integrated AM platforms. This paper provides a comprehensive overview of the current state of MMGAM, structured around four key themes: (i) key material properties and interfacial compatibility considerations, (ii) AM techniques adapted for multi-material glass systems, (iii) material combinations and strategies for functional integration, and (iv) future challenges and research directions. Particular emphasis is placed on the challenges of interfacial bonding, material compatibility, and the formulation of printable glass systems suitable for co-processing with other materials. The review explores recent strategies for enabling multifunctional, integrated architectures that combine optical, structural, or chemical functionalities within a single component. By synthesising current advances and outlining key research gaps, this work aims to accelerate the development and broader adoption of MMGAM technologies.

Keywords Additive manufacturing · Glass · Multi-material · Silicon dioxide · 3D printing

1 Introduction

Glass is a technologically vital material, widely used in scientific research, industry, and everyday applications. Its exceptional properties including optical transparency, mechanical strength, chemical durability, thermal resistance, and electrical insulation, make it indispensable across disciplines such as chemistry, biology, optics, and optoelectronics [1–4]. High-purity glasses like fused silica are particularly valued in photonics and microfluidics, but are notoriously difficult to process due to their high melting temperatures and brittleness [5]. Conventional manufacturing methods such as high-temperature casting and wet etching offer limited geometric flexibility and often rely on hazardous chemicals [5, 6]. Although polishing can improve surface quality, intricate or micro scale features are prone to damage during post-processing [7].

Additive manufacturing (AM), also known as 3D printing, enables the layer-by-layer fabrication of complex

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geometries from digital models. It offers transformative advantages including design freedom, rapid prototyping, reduced material waste, and on-demand customisation [8–11]. While AM is well established for polymers, metals, and ceramics, extending it to glass has been challenging due to the material's high viscosity, thermal sensitivity, and limited availability of printable feedstocks. Nonetheless, recent advances have demonstrated the feasibility of glass AM across multiple techniques, alongside the development of new machinery and equipment [12, 13104]. In parallel, material extrusion-based multi-material printing has rapidly advanced via multi-nozzle deposition, material switching, and in-situ mixing strategies [14]. These developments position glass AM as a frontier domain for applications in optics, microfluidics, sensing, and harsh-environment devices.

Within this evolving landscape, multi-material glass additive manufacturing (MMGAM) has emerged as a promising next step. In this work, MMGAM denotes AM processes in which at least one constituent is a glass and spatially distinct regions differ in composition and/or material class. Furthermore, MMGAM is divided into two subclasses: (i) glass-glass multi-material systems, where two or more glass compositions are combined within a single part, either within one family or across different families (e.g. silicate, borosilicate, phosphate). These systems include compositional or functional gradients (e.g. refractive-index tuning or ion-exchangeable layers) and configurations that exploit structural or thermophysical contrasts [15–17]; (ii) hybrid systems, in which glass is combined with at least one other material class (e.g. polymers, metals, or ceramics, including glass-ceramic combinations), to provide mechanical reinforcement, electrical functionality, or packaging features [18–20].

Adopting a glass-centred perspective, MMGAM is surveyed across both subclasses, highlighting shared processing constraints alongside subclass-specific opportunities. This multi-material design freedom enables spatial control over composition and local functionality, opening the door to applications such as gradient-index optics, lab-on-glass microdevices, and hybrid microfluidic-photonic systems.

Several recent reviews have advanced the understanding of glass and multi-material AM. Hasanov et al. [21] reviewed broader trends in multi-material AM, emphasising interfacial bonding and process compatibility challenges. Shaw et al. [22] critically examined AM techniques for glass across scales, highlighting process limitations and semi-solid fabrication approaches. Scheithauer et al. [23] reviewed recent developments in ceramic-based multi-material AM, highlighting functionally graded concepts and the central role of debinding/co-sintering constraints that are also instructive for glass-ceramic MMGAM. Barbera et al. [16] demonstrated multi-material volumetric printing of silica-based glasses with tunable porosity and composition. Chen et al. [24] focused on photopolymerisation-based AM for optical elements, while Arriaga-Dávila et al. [25] explored micro-optical glass-ceramic architectures via two-photon lithography. However, none of these reviews fully addressed MMGAM, particularly regarding integration strategies, glass formulation, and cross-scale functionality.

MMGAM presents significant technical challenges because differences in thermal expansion, viscosity-temperature behaviour, sintering shrinkage, and chemical reactivity across interfaces can drive delamination, residual stress, devitrification, and porosity. Addressing these issues requires coordinated advances in glass chemistry, interface design, and process control, supported by a process–structure–property perspective for multi-material builds. MMGAM therefore lies at the intersection of glass AM and multi-material printing, motivating an integrated review of formulation, interfacial compatibility, and suitable AM routes.

Accordingly, this review is structured around four core objectives: (i) evaluating glass materials and their printability with a focus on thermophysical and interfacial constraints; (ii) reviewing AM techniques suitable for multi-material glass systems; (iii) surveying emerging MMGAM applications; and (iv) identifying challenges and opportunities for scaling MMGAM toward broader scientific and industrial use. Figure 1 summarises the MMGAM landscape as a conceptual roadmap that links material selection, processing strategies, integration modes, and application domains.

Driven by innovations in both glass chemistry and AM hardware, glass AM is receiving growing attention across sectors including optics, microelectronics, sensors, and biomedical systems. This increasing interest is reflected in publication trends. Figure 2 presents a bibliometric analysis based on Scopus data. The first dataset was obtained using the keywords: *"additive manufacturing" OR "3D printing" AND "glass" AND NOT "glass transition" AND NOT "glass transient"*, which reveals a sharp increase in publications since 2015 (Fig 2a), with leading contributions from China, the United States, and Germany, followed by the United Kingdom, India, and Italy (Fig 2b). This confirms that glass AM has become an emerging and rapidly growing research field.

To further refine the scope, a second dataset was extracted using the terms: *"additive manufacturing" OR "3D printing" AND "multi-material" AND "glass" AND NOT "glass transition" AND NOT "glass transient"*. Although the number of publications is still relatively small, the trend shows steady growth (Fig 2c), indicating that MMGAM is gaining increasing attention as a specialised research direction. This distinction highlights how MMGAM is emerging from the

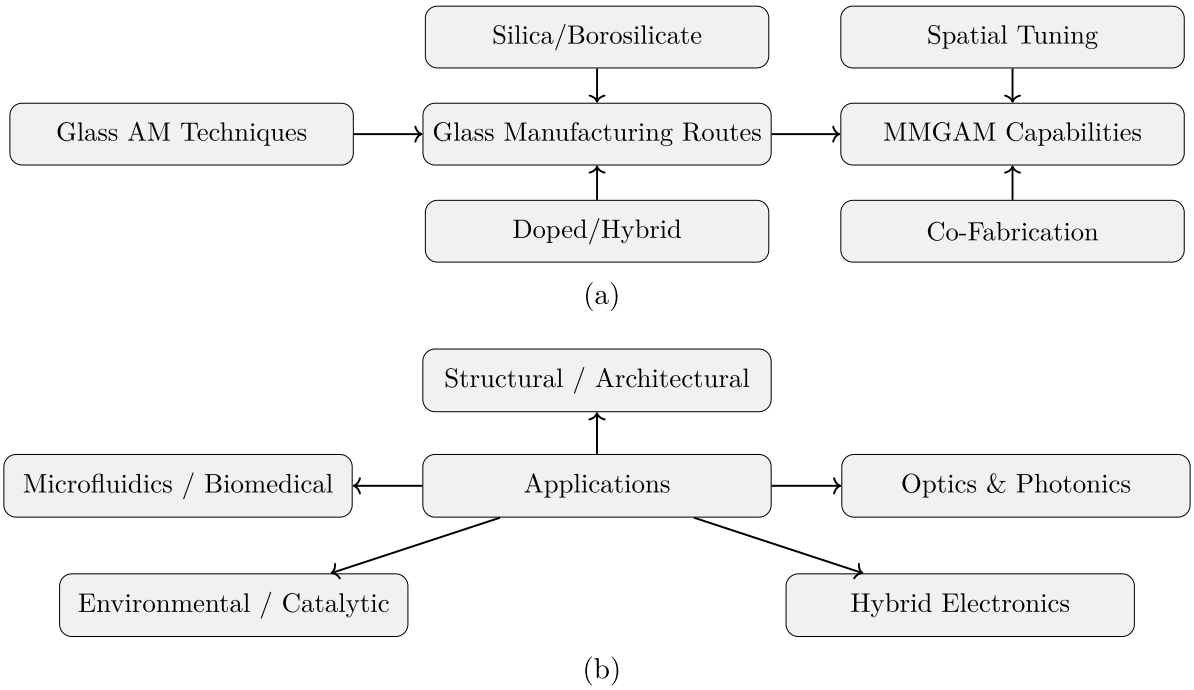


Fig. 1 MMGAM workflow: **a** AM techniques and glass classes enable spatial tuning and cofabrication; **b** application domains

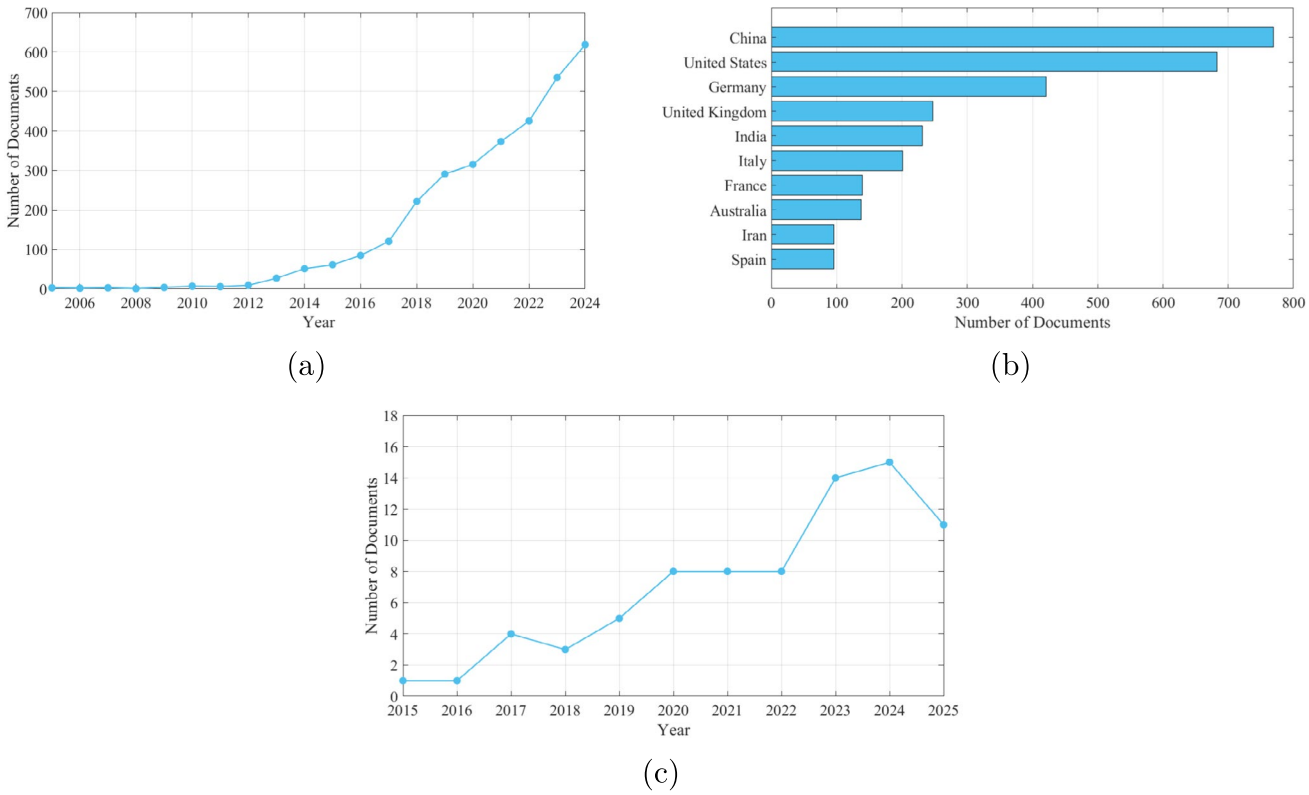


Fig. 2 Statistics values of AM-glass academic publications, **a** over year and, **b** pioneering countries between 2005–2024, **c** MMGAM over year (source: Scopus)

broader glass AM domain, driven by the demand for hybrid devices and multifunctional glass-based systems.

Despite this progress, focused overviews of MMGAM are still limited. This review therefore compiles the main

glass types and properties relevant for AM, surveys AM techniques that have been adapted to multi-material glass systems, and summarises reported material combinations, application examples, and remaining challenges. With this rising interest in glass AM technologies, a key consideration is the suitability and adaptability of different glass formulations for AM. The following section therefore discusses the types of glass materials employed in AM and the key properties that govern their printability.

2 Glass materials and their printability in AM

Glass materials used in AM offer a distinctive combination of beneficial properties, including optical transparency, thermal and chemical stability, and mechanical durability. These attributes support a wide range of functional applications, particularly in optics, electronics, and high-temperature environments. However, the same properties present processing challenges due to high melting points, high viscosity, and intrinsic brittleness.

In AM contexts, glass shares rheological and sintering characteristics with ceramics. Transparency is particularly valued in photonic and optical applications, where optical clarity is critical [26]. Its high-temperature resistance makes it suitable for extreme environments, while chemical inertness and mechanical strength contribute to long-term performance. Nevertheless, the inherent brittleness of glass increases the risk of fracture during deposition and cooling, while its high viscosity and demanding thermal requirements limit design resolution and process flexibility. At the component scale, these effects lead to warpage and residual stress, interlayer debonding, porosity (entrapped bubbles), devitrification, and increased surface roughness, compromising dimensional accuracy and optical quality; in multi-material builds, coefficient of thermal expansion (CTE) mismatch further weakens interfacial adhesion and reliability.

2.1 Common glass types used in AM

The printability and post-processing behaviour of glass are governed largely by its chemical composition. The following types are commonly explored in AM research and applications:

- *Fused silica (quartz)* offers excellent optical clarity and thermal stability, with an ultra-low thermal expansion coefficient ($0.5–0.6\times10^{-6}$/K), making it ideal for precision applications. However, its softening point exceeds 1500 °C, requiring advanced thermal systems.
- *Borosilicate glass* known for its chemical durability and thermal stability. With a lower thermal expansion coefficient (3.3×10^{-6}/K) and a working point around 1200 °C, it is widely used in laboratory and industrial contexts.
- *Aluminosilicate glass* combines high strength, chemical resistance, and thermal stability. Commonly used in display technologies and sensor substrates. Its working point exceeds 1300 °C, with a strain point near 666 °C.
- *Soda-lime silica glass* the most commercially prevalent glass type, used in windows and containers. It is cost-effective and easy to form but has a high thermal expansion coefficient (9.1×10^{-6}/K), limiting thermal shock resistance.
- *Germanate glass* characterised by a high refractive index and strong infrared (IR) transmission. Commercial examples like Corning 9754 show good environmental durability and processability (softening point 874 °C), making them suitable for photonic and sensing applications using laser-based AM.

These compositional profiles are summarised in Table 1, which compiles representative commercial glass types relevant to AM and details their oxide compositions, and further summarises their printability status and application domains.

2.2 Key physical and mechanical properties of glass

Understanding the physical and mechanical behaviour of glass is critical to its performance in AM. Properties such as Young's modulus, density, Poisson's ratio, thermal conductivity, and electrical resistivity influence how glass behaves during printing, cooling, and service. A comparison of physical and mechanical properties, including thermal conductivity, expansion, and electrical resistivity, is provided in Table 2.

The CTE is especially important, as it affects dimensional stability and thermal stress. Low-CTE glasses such as fused silica and borosilicate are preferred for minimizing warping and cracking. Germanate glasses offer additional functionality, including high refractive index and IR transparency, with relatively moderate softening temperatures that support laser-assisted AM.

2.3 Viscosity behaviour and processing temperatures

Viscosity governs the flow behaviour, interlayer bonding, and sintering dynamics of glass during AM. As glass is heated, it softens gradually rather than melting abruptly, and its viscosity decreases exponentially with temperature.

Table 1 Representative glass compositions, characteristics, and AM relevance (adapted from [27–34])

Glass type	Commercial name	Composition (wt%)	Characteristics	Printability (current status)	AM-relevant applications
Soda-lime	SCHOTT AR-GLAS®HIL 0100	SiO_2 69–71, Na_2O 13–14, CaO 5, Al_2O_3 4, K_2O 3, MgO 3	Low cost, easy to form and process, moderate chemical durability	Limited / demonstrated in research	Structural or decorative components via paste-/powder-based AM; sintered parts and prototypes (non-optical grade)
Borosilicate	DURAN®	SiO_2∼81, B_2O_3 ∼13, Na_2O ∼4, Al_2O_3∼2	High thermal shock resistance, good chemical durability	Feasible / widely used in AM studies	Direct-melt extruded objects, microfluidic chips, thermally robust labware geometries
Aluminosilicate	Corning 1737F	SiO_2 55–60, Al_2O_3∼21, BaO/CaO ∼12, others (SrO, TiO_2)	High strength, high thermal resistance, good electrical insulation	Challenging / specialised set-ups required	High-temperature functional substrates and coatings; exploratory AM demonstrators
Fused silica	Corning HPFS® 7980	SiO_2 > 99.99	Very high purity, excellent UV transparency, high thermal stability	Widely demonstrated / mature in research	Optical micro-lenses, waveguides, micro-optical/mechanical devices, transparent lattice architectures
Germanate	Corning 9754	GeO_2-rich, low alkali	High refractive index, IR transmission up to 5.3 µm, moderate softening temperature	Emerging / limited demonstrations	IR-transparent components and photonic structures; niche AM prototypes

Table 2 Physical and mechanical properties of selected glasses (adapted from [27–32, 35])

Property	Soda-lime (HIL 0100)	Borosilicate (DURAN®)	Alumino-silicate (1737F)	Fused silica (HPFS® 7980)	Germanate (corning 9754)
Density (g/cm^3)	2.50	2.23	2.54	2.20	3.51
Young's modulus (GPa)	72	63	70.9	73	84.1
Poisson's ratio	0.22	0.20	0.23	0.16	0.29
Thermal conductivity (W/m.K)	1.1	1.2	1.1−1.2	1.38	0.42
Thermal expansion (10^{-6}/K)	9.1	3.3	4.2	0.5−0.57	7.5
Electrical resistivity (Ω·cm) @ 250° C	10^7	10^{13}–10^{14}	$10^{13.5}$	$>10^{16}$	10^{12}–10^{13}

Table 3 Standard viscosity reference points for key commercial glasses (compiled from [27–30, 32, 35])

Reference point (Pa.s)	Soda-lime (HIL 0100)	Borosilicate (DURAN®)	Alumino-silicate(1737F)	Fused silica (HPFS® 7980)	Germanate (corning 9754)
Working point (10^4)	1040	1260	1312	>1800	1050
Softening point ($10^{7.5}$)	720	825	975	1585	874
Annealing point (10^{13})	530	560	721	1042	735
Strain point ($10^{14.5}$)	495	518	666	893	697

All temperatures are in $^\circ C$

This thermal behaviour is typically characterised by four standard reference temperatures.

The *working point* (∼10^4 Pa.s) defines the condition where glass flows readily and can be shaped or deposited. As the temperature decreases, the *softening point* (∼$10^{7.5}$ Pa.s) marks the transition where the glass begins to deform under its own weight. Further cooling reaches the *annealing point* (∼10^{13} Pa.s), a temperature range where internal stresses can be relieved gradually through structural relaxation. Below this lies the *strain point* (∼$10^{14.5}$ Pa.s), where the glass becomes effectively rigid and any residual stresses become locked in. These reference points vary among glass types and define the operating temperature range for AM. Table 3 summarises these key temperatures for selected commercial glasses. Accurate control of these setpoints is essential for part quality. Misalignment can degrade optical, mechanical, and dimensional performance by inducing defects, residual stresses, and loss of transparency.

2.4 Glass formulations for multi-material AM

Recent advances in MMGAM rely on two complementary fronts: the development of printable functionally tunable glass formulations, and the co-fabrication of glass with

dissimilar materials to create hybrid systems with integrated functionality.

Formulation innovations have focused on improving printability, compatibility, and performance across AM platforms. Nanoparticle-based suspensions and photoresins have enabled more uniform sintering and shape fidelity, while metal-doped glasses offer tunable optical, electrical, or catalytic properties. Graded or hybrid compositions are increasingly used to mitigate thermal mismatch and enhance interfacial bonding. Notable examples include silica-titania inks engineered for gradient-index optics [15], metal oxide-doped silica resins for digital light processing [36], and photo-cross-linkable silica suspensions for fast sintering with high optical quality [37].

In parallel, multi-material integration strategies have advanced the functional scope of MMGAM by enabling the combination of glass with polymers, metals, and ceramics within a single build. Approaches can be broadly categorised as (i) modified glass systems incorporating dopants, fillers, or reinforcements, and (ii) co-fabrication processes that deposit multiple distinct materials simultaneously or sequentially. In MMGAM, compositional modifiers (dopants/oxides) can also shift viscosity-temperature behaviour and thermal-expansion response, narrowing the co-processing range and increasing interfacial-stress risk [22]. Multicomponent or dopant-loaded resins can alter photopolymerisation/phase-separation behaviour and the microstructure formed after heat treatment, affecting porosity and bonding between regions [16, 38, 39].

For instance, Yang et al. [18] demonstrated multi-photon nanoprinting of hybrid glass-ceramic-polymer microstructures, while Cheng et al. [19] developed glass-ceramic blends with enhanced laser absorption for powder-based AM. Hisham et al. [20] produced dye-infused, multi-material contact lenses, and Ouyang et al. [39] fabricated rare-earth-doped luminescent glass via UV curing and femtosecond laser structuring. Additional examples include conductive glass-ceramic systems using silver inks [40] and laser-bonded metal-glass interfaces [41]. Low-temperature hybrid MMGAM is also possible via polymer-glass composites; for example, Roitero et al. [42] used Arburg Plastic Freeforming to print P(L,DL)LA filled with bioactive glass and discussed degradation-related reproducibility limits relevant to implantable devices. Bioactive-glass formulations are particularly sensitive to ink design and the debinding/sintering schedule, and in MMGAM they require shrinkage/CTE-aware interface design when combined with inert or structural phases [7, 22, 43, 44].

These developments have opened pathways for next-generation applications, including photonic components with spatially varying refractive indices, biomedical implants with bioinert or bioactive surfaces, and multi-functional microelectronic packaging. Materials such as chalcogenide glasses, PbSe, and $LiNbO_3$, known for their high refractive indices, have been successfully fabricated into 3D structures via direct multi-photon laser printing [18]. In addition, silica nanopowders embedded in photoresists have enabled high-index-contrast structures for waveguides and optical sensors [45]. Recent progress in sol–gel-derived glass ceramics has enabled the first robocasting of bioactive glass scaffolds with optimised rheology. The printed structures exhibited good mechanical strength and bioactivity, indicating strong potential for tissue engineering [43].

Nonetheless, challenges remain in achieving seamless integration due to disparities in thermal expansion, sintering kinetics, and interfacial chemistry. Recent research is increasingly turning to hybrid AM platforms and graded interface design to overcome these limitations and realise robust, application-specific multi-material glass architectures. Having examined the physical and chemical characteristics of glasses used in AM, the following section focuses on the specific AM methods that enable their processing, particularly in multi-material contexts.

3 AM approaches for multi-material glass integration

MMGAM has emerged as a promising frontier in advanced fabrication, enabling not only the spatial tuning of glass composition, dopants, and functionalities, but also the co-fabrication of glass with dissimilar materials, such as polymers, metals, and ceramics, within a single integrated structure. These capabilities offer unprecedented design freedom but also introduce significant challenges related to thermal compatibility, rheological behaviour, interfacial bonding, and deposition sequencing. As a result, only a subset of AM techniques have been adapted to support MMGAM, often requiring specialised hardware configurations and tailored processing strategies.

This section reviews AM processes relevant to MMGAM, emphasising their suitability, limitations, and demonstrated capabilities for spatially and compositionally complex systems. The most commonly employed techniques for MMGAM correspond to the standardised process families defined in ISO/ASTM 52900:2021 [46], namely material extrusion with thermally or chemically reactive bonding (MEX; including thermally reactive bonding-MEX-TRB, and chemically reactive bonding-MEX-CRB), vat photopolymerisation (VPP; encompassing ultraviolet laser beam-UVL, mask projection-UVM, and LED-based curing-LED), powder bed fusion (PBF; comprising both laser-beam PBF

for metals-PBF-LB/M, and laser-beam PBF for polymers, ceramics, and glass-PBF-LB/P), and directed energy deposition (DED). Figure 3 illustrates representative examples of glass components fabricated using the main AM process families, highlighting the diversity of achievable geometries, optical qualities, and application potential across different process routes.

3.1 Material extrusion (MEX)

MEX methods for glass AM can be broadly categorised into two types: MEX-CRB, encompassing paste-based direct ink writing (DIW) using glass-polymer or glass-particle feedstocks; and MEX-TRB, including direct melt extrusion and filament-fed deposition. DIW utilises shear-thinning or particle-rich inks composed of glass powders, polymeric binders, and functional additives. These inks are extruded at room or moderate temperatures and require sintering to remove organics and achieve full densification. In contrast, direct melt and filament-fed techniques heat glass rods or filaments above 1000 °C to enable continuous, binder-free deposition, producing transparent or fully dense glass structures without additional post-processing.

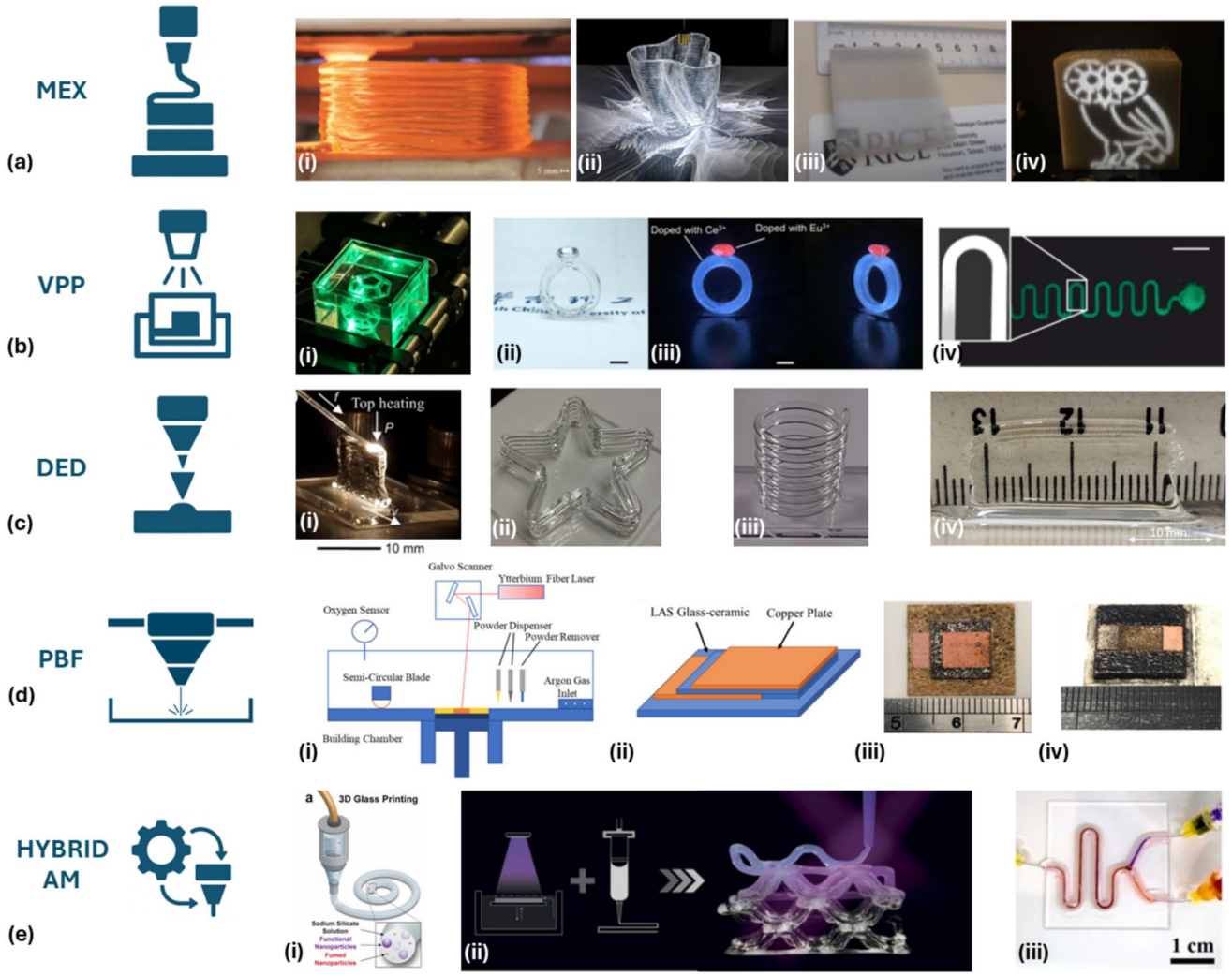


Fig. 3 Representative glass-AM process families and example components. **a** MEX: (i) molten glass deposition (adapted from [7]), (ii) architectural artefact (adapted from [7]), (iii–iv) optical components (adapted from [48]). **b** VPP: (i) TPP microstructure (adapted from [45]), (ii–iii) doped silica glass with engineered emission colours (Ce^{3+}/Eu^{3+}) (adapted from [39]), (iv) suspended hollow microstructures (adapted from [49]). **c** DED: (i) filament-fed fused-quartz printing with CO_2-laser heating (adapted from [104]), (ii) star-shaped structure (adapted from [51]), (iii) helical spring (adapted from [51]), (iv) single-wall print (adapted from [50]). **d** PBF: (i) multi-material LPBF setup, (ii) parallel-plate capacitor schematic, (iii) printed copper/lithium aluminosilicate, (iv) printed copper stainless steel/lithium aluminosilicate (i-iv are adapted from [19]). **e** Hybrid-AM: (i) low-temperature printing using metastable silicates (adapted from [44]), (ii) hybrid fabrication of transparent silica structures (adapted from [53]), (iii) printed glass microfluidic channels with ink injection (adapted from [52])

A variety of MEX-based approaches have been demonstrated across this temperature spectrum. Low-temperature strategies include sodium silicate-based DIW inks for functional glass composites [44], silica nanoparticle inks for transparent optics [54], and robocasting of silicate pastes with tailored rheology and sintering profiles [55]. These methods offer material versatility and compositional tunability but rely on precise post-processing.

Intermediate-temperature extrusion using thermoplastic nanocomposite filaments [56] enables printing on filament-fed MEX systems at 160–190 °C. More generally, multi-material polymer MEX offers useful laminate design rules (near rule-of-mixtures scaling in tension and shell-dominated bending response), which are informative for MMGAM architectures such as shell-core or layered builds [57]. At the high end, molten glass extrusion (MEX-TRB) reaches temperatures exceeding 1000 °C. The G3DP platform developed by Klein et al. [47] exemplifies this approach, extruding soda-lime glass through an alumina-zirconia-silica nozzle to produce dense, transparent components without sintering.

In the context of MMGAM, DIW and paste extrusion offer key advantages due to their room-temperature processing. These methods enable integration of multiple glass compositions with distinct dopants or functionalities, allowing spatial control through programmable nozzle switching or multi-feed systems. However, the required high-temperature sintering remains a major limitation, especially for co-processing with thermally sensitive materials such as polymers, certain metals, or bioactive agents.

Direct melt and filament-fed systems, while restricted by extreme processing conditions, offer MMGAM potential through multi-nozzle setups or sequential feeding of distinct glass rods. They can also accommodate other high-temperature-compatible materials, such as metals, ceramics, or engineering plastics, enabling fabrication of gradient-index optics, functional interfaces, and composite structures. Nonetheless, these approaches face challenges in achieving compositional compatibility due to thermal expansion mismatch, interfacial bonding issues, and viscosity disparities at elevated temperatures.

MEX-based glass printing commonly encounters porosity, cracking, dimensional inaccuracy, and shrinkage, mainly arising from incomplete densification, thermal gradients, and material incompatibility. These are among the most frequently reported challenges affecting the optical and structural quality of printed glass components.

In MEX-based MMGAM, interfacial quality in MEX-CRB (DIW and paste extrusion) is largely governed by shrinkage matching and filament coalescence, so well-tuned rheology, solids loading and drying/sintering schedules are needed to avoid pore chains and microcracks along material boundaries [7, 43, 44, 54, 55]. In MEX-TRB (direct-melt and filament-fed schemes), high-temperature wetting, viscosity compatibility, CTE matching and appropriate cooling/annealing link the processing conditions directly to interlayer bonding, bubble content and residual stress, and therefore to transparency and mechanical reliability of the printed glass parts [22, 47, 56, 58].

3.2 Vat photopolymerisation (VPP)

VPP techniques such as stereolithography (SLA), digital light processing (DLP), and two-photon polymerisation (TPP) have enabled the fabrication of complex, high-resolution glass structures using nanoparticle- or precursor-loaded resins, followed by thermal debinding and sintering above 1000 °C [5]. These methods support excellent geometric precision, optical transparency, and functional doping, making them well-suited for photonic and micro-optical applications. In the context of MMGAM, VPP holds promise through spatially resolved compositional control, such as multicolour luminescent glass using dopant-loaded SLA [59, 60], or space-selective doping in silica glass via stereolithography and femtosecond laser-assisted techniques [39]. TPP has also demonstrated the fabrication of photonic microstructures with rare-earth doping for tunable optical responses [61]. Volumetric printing offers additional capabilities, enabling voxel-level resolution for complex architectures, multicolour structuring [16], and microfluidic device fabrication in silica glass [62, 63]. However, the reliance on single-resin chemical formulation and high-temperature sintering remains a limitation, especially for integrating dissimilar or thermally sensitive materials.

To expand the applicability of VPP for MMGAM, several strategies have been explored. Multi-resin printing platforms, using automated vat switching or multi-pattern projection, can spatially deposit distinct glass precursors or dopant formulations. Graded composition can be achieved via grayscale exposure or continuous resin mixing during printing. In two-photon and volumetric lithography, voxel-level control enables fine spatial tuning of refractive index and luminescent properties [16, 39]. Additionally, photoresponsive or wavelength-selective photoinitiators can enable selective polymerisation within mixed-resin environments. Despite these advances, MMGAM via VPP remains constrained by interfacial diffusion during sintering, resin compatibility issues, and challenges in achieving sharp material transitions without phase separation or print artifacts.

Moore et al. [38] introduced a digital light-processing approach that exploits photopolymerisation-induced phase separation in hybrid resins to produce dense, multicomponent oxide glasses with complex geometries and tunable porosity. This work established a foundation for chemical

versatility and compositional control in glass VPP. Building on such photopolymer-based hybrid glass AM concepts, Hatscher et al. [64] demonstrated the additive fabrication of fused silica microstructured optical fibers from a nanocomposite resin, highlighting the precision and optical quality achievable through 3D printing of silica networks.

In VPP-based MMGAM, interfacial bonding is largely dictated by resin compatibility: differences in polymerisation behaviour, shrinkage and ceramic yield between doped and undoped glass-forming resins can generate porosity, residual stresses and composition gradients at material boundaries [16, 38]. Using a common silica backbone, miscible resin chemistries and co-optimised debinding/sintering schedules has been shown to produce nearly continuous optical interfaces and crack-free multi-material parts [16, 38, 39]. Resin formulation (nanoparticle loading, precursor and photoinitiator chemistry) and exposure conditions determine green-body packing and crosslink density, while the debinding/sintering profile controls densification, bubble formation and crystallisation, so that processing conditions are directly reflected in microstructure, refractive index and luminescent response, and thus the optical and mechanical reliability of the printed glass [5, 38, 39, 62].

3.3 Directed energy deposition (DED)

DED has recently emerged as a viable method for glass AM, though it is more traditionally associated with metals using arc[65], laser, or electron beam[66] with powder or wire as the raw material. In glass DED, a focused laser; typically a CO_2 source; is used to locally melt glass feedstock delivered in powder or filament form. Among the two variants, wire-based DED (DED-W) and powder-based DED (DED-P), the DED-W has shown greater promise for MMGAM due to its superior thermal control, smoother deposition, and full material utilisation [50, 67, 68]. Glass filaments ranging from 0.125 to 2 mm in diameter can be fed continuously to produce dense, partially transparent structures with minimal porosity.

Examples include the fabrication of silica optics [68], borosilicate devices [69], and rare-earth-doped photonic elements [70]. Fröhlich et al. [71] demonstrated CO_2-laser-based rod-fed deposition of borosilicate and soda-lime glass, achieving stable near-net-shape walls with reduced residual stress through optimised process parameters, highlighting the scalability of filament-fed DED for large glass components.

A notable advancement is the digital glassforming method by Deutsch-Garcia et al. [72, 73], which processes soda-lime glass rods using volumetric laser heating via dopant-tuned optical penetration depth. This enables improved deposition rates, reduced bubble formation, and compositional control, key benefits for MMGAM. In contrast, powder-fed DED faces significant limitations due to poor powder flowability, thermal instability, and residual porosity. While some success has been achieved using soda-lime-silica powders [74], the technique remains less mature than its filament-based counterpart for glass.

From an MMGAM perspective, DED offers potential through multi-filament feeds or coaxial wire integration, enabling spatial control of composition or dopant loading. High-temperature compatibility also allows co-processing with certain metals and ceramics. However, challenges such as thermal expansion mismatch, laser-material coupling disparities, and interfacial cracking limit the integration of dissimilar materials. Furthermore, precise spatial switching between glass types or compositions remains technically complex.

In glass DED, interfaces between different glasses are strongly governed by melt-pool dynamics and sharp thermal gradients; mismatches in thermal expansion or absorption behaviour readily promote microcracking and delamination [68, 71]. Substrate preheating, optimised scan paths and, in some cases, compositional grading have been used to stabilise walls and graded regions, but fully defect-free integration of strongly contrasting materials remains rare [71, 72]. Process parameters (laser power/scan speed/beam path, feed rate and preheating) control melt-pool size and cooling rate, and thus residual stress, porosity and surface waviness, which directly impact transparency and strength; especially at multi-material interfaces [22, 72]. Overall, while DED is conceptually attractive for MMGAM via feedstock switching, it remains at an early development stage for glass even in single-material builds, and most demonstrations are still proof-of-concept.

3.4 Powder bed fusion (PBF)

Powder bed-based techniques, including selective laser melting (SLM) and selective laser sintering (SLS), have been explored only marginally for glass and glass-ceramic systems. Early studies demonstrated the feasibility of processing soda-lime, fused silica, and lithium aluminosilicate (LAS) glasses using laser PBF [75–77], but the resulting parts often exhibited high porosity, thermal cracking, and limited optical quality due to poor laser absorptivity and sharp thermal gradients. Here, PBF-LB/P denotes laser-based powder-bed processing for polymers/ceramics/glass; for glass(-ceramic) systems it is most often reported as SLS or laser-assisted densification rather than full melting (LPBF).

More recently, Cheng et al. [19] fabricated LAS glass-ceramic/metal components via multi-material PBF-LB, illustrating the potential of combining structural and

conductive phases within a single build. Zocca et al. [78] demonstrated selective laser densification of LAS glass-ceramic tapes produced by layer-wise slurry casting, while Xiao et al. [79] developed an indirect SLS route for apatite-wollastonite (A-W) glass-ceramics, yielding porous, bioactive scaffolds suitable for bone implant applications. PBF-LB of quartz glass has also been advanced by introducing grooved 316 L substrates and height-adaptive laser power strategies, which form a mixed glass-metal transition layer, improve bonding, and allow control over porosity to realise optically non-uniform quartz components with distinct top and bottom surface responses [80].

Compared to extrusion- or resin-based AM routes, PBF offers high spatial resolution and efficient powder recyclability, making it attractive for structural and functional oxide systems. However, its application to MMGAM remains limited by weak laser-glass coupling, residual stress, and poor interfacial bonding between dissimilar phases. Consequently, PBF currently serves mainly as an experimental platform for glass-ceramic or glass-metal composites rather than a practical route for transparent multi-material glass manufacturing. [80]

In PBF of glass and glass-ceramic systems, limited laser absorption and a narrow processing range make interfacial bonding fragile: small changes in energy density or preheating can shift builds from insufficient necking to microcracked or partially crystallised interfaces, especially in multi-material parts [75, 77]. Despite these limitations for transparent, crack-free bulk glass, PBF-LB offers exceptional geometric freedom for porous and lattice architectures, and it is among the few glass-AM routes where mechanical testing has been reported alongside demonstrations of complex shapes [77]. However, current glass-metal or glass-ceramic MMGAM via PBF remains largely proof-of-concept and rarely achieves transparent or highly reliable interfaces [19]. Overall, present implementations favour opaque scaffolds or infiltrated composites rather than fully dense, optically clear MMGAM components [19, 77, 78].

3.5 Hybrid-AM

Hybrid AM is an emerging approach across various material domains. A notable example is the hybrid metal multi-station, multi-axis system introduced by Karunakaran et al. [81]. In general, Hybrid AM integrates multiple AM processes or couples AM with complementary fabrication techniques to improve resolution, multi-material compatibility [82, 83], residual-stress management [84], and production efficiency. These concepts are particularly relevant for MMGAM, where no single AM process can simultaneously satisfy material, thermal, and geometric constraints. Recent Hybrid-AM platforms for MMGAM include:

- UV-assisted DIW enabling simultaneous deposition and in-situ curing of silica-polymer inks [53],
- volumetric printing systems capable of synchronised multi-material photopolymerisation [16],
- room-temperature imprinting combined with extrusion overprinting for microfluidic glass structures [85],
- laser-assisted extrusion combining paste deposition with in-situ CO_2-laser melting for transparent fused silica fabrication [52].

These strategies broaden the accessible process space for MMGAM, but they also introduce challenges in synchronisation, rheological matching, and thermal management. Reported examples span both (i) hybridisation during shaping (e.g., imprint-extrusion and UV-assisted DIW for stabilizing geometry and interfaces prior to conversion) and (ii) localised consolidation/finishing (e.g., laser-assisted fusion applied only where needed). For instance, Kotz et al. [49] used sacrificial templating to produce hollow microstructures in fused silica, while Haq et al. [85] demonstrated a hybrid imprint-extrusion approach to pattern microfluidic glass channels at room temperature with water-based particulate inks. Early work by Zhang et al. [86] combined powder-based 3D printing of alumina preforms with glass infiltration to form alumina-glass composites, representing an early hybrid glass-ceramic route and a precursor to modern multi-material glass printing.

Beyond combining processes during fabrication, Hybrid-AM can also be sequential, coupling additive shaping with subsequent transformation stages to achieve final glass properties. Annealing, polishing, or thermal conversions are common examples. Gailevičius et al. [17] demonstrated such a sequential hybrid approach, employing multiphoton lithography followed by nanoscale heat treatment to convert printed hybrid structures into fully inorganic glass-ceramics with enhanced optical and structural performance.

In MMGAM, the central technical question is how to design and consolidate interfaces between glass and secondary phases without damaging previously printed regions. Across hybrid concepts, partial consolidation, local densification, and final heat treatment together define the local thermal history at and near interfaces, and thus influence defect evolution, residual stress, and interface reliability [17, 49, 52, 53, 85]. Hybrid-AM therefore provides additional process degrees of freedom for managing interfaces in MMGAM, but it can also increase path-dependence and sensitivity to misalignment or mismatched thermal cycles [52, 53, 85].

A practical hardware model for high-temperature co-fabrication is a dual-feed MEX-TRB configuration (Fig. 4), where molten glass is deposited alongside a secondary material using independent feeding and extrusion units

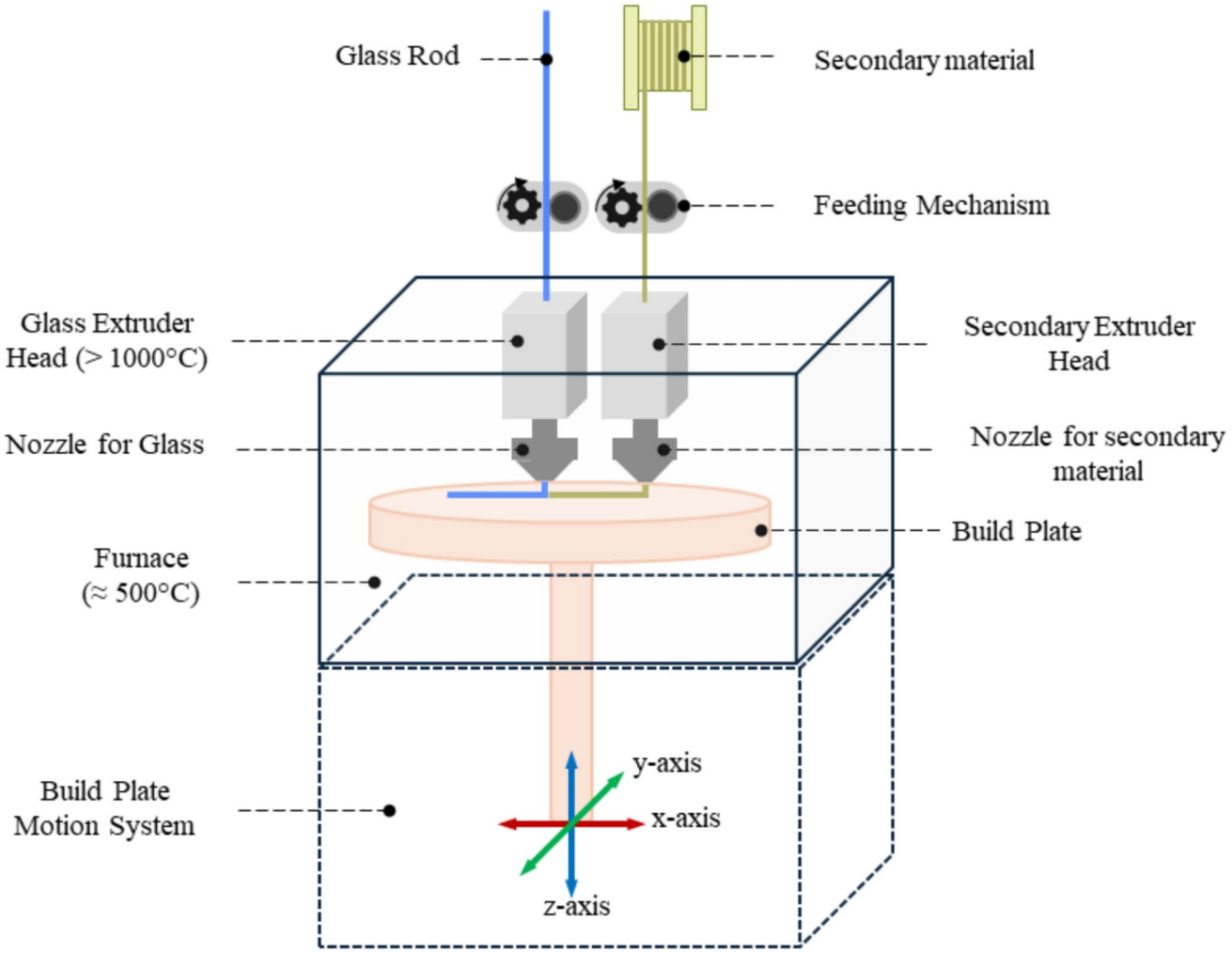


Fig. 4 Conceptual schematic of a hybrid MEX-TRB system for MMGAM, showing dual extrusion heads for glass and a secondary material within a controlled thermal chamber

in a controlled thermal enclosure. Key challenges include achieving precise thermal control for both extrusion mechanisms, maintaining an appropriate chamber temperature, managing synchronised material-switching, and ensuring strong interfacial bonding (or controlled delamination) between dissimilar materials after printing.

To place Hybrid-AM in the broader multi-material landscape, Fig. 5 summarises the interrelation of glass with major AM material families (polymers, metals, and ceramics), highlighting typical feedstock forms that enable practical hybridisation routes. These intersections point to emerging strategies where glass can be co-processed with polymers, ceramics, or metals to realise composite structures, functional interfaces, and thermally graded components.

Despite its versatility, Hybrid-AM remains constrained by process complexity, thermal mismatch, and alignment inaccuracies, which can lead to interfacial defects and reproducibility issues. Achieving stable integration of multiple processes and materials is still a key challenge for scalable MMGAM implementation.

3.6 Comparative suitability of AM techniques for MMGAM

To complement the process-specific discussions above, Table 4 summarises the types of glass materials successfully demonstrated across various AM platforms.

MMGAM presents transformative opportunities for fabricating glass components with spatially varying compositions, functional gradients, and integrated material interfaces. Realizing this potential requires AM strategies that enable fine control over feedstock delivery, thermal gradients, and interfacial bonding.

Among the reviewed techniques, extrusion-based methods, particularly DIW and melt extrusion, and VPP stand out as the most promising routes for MMGAM. These methods enable multi-feed deposition, dopant modulation, and

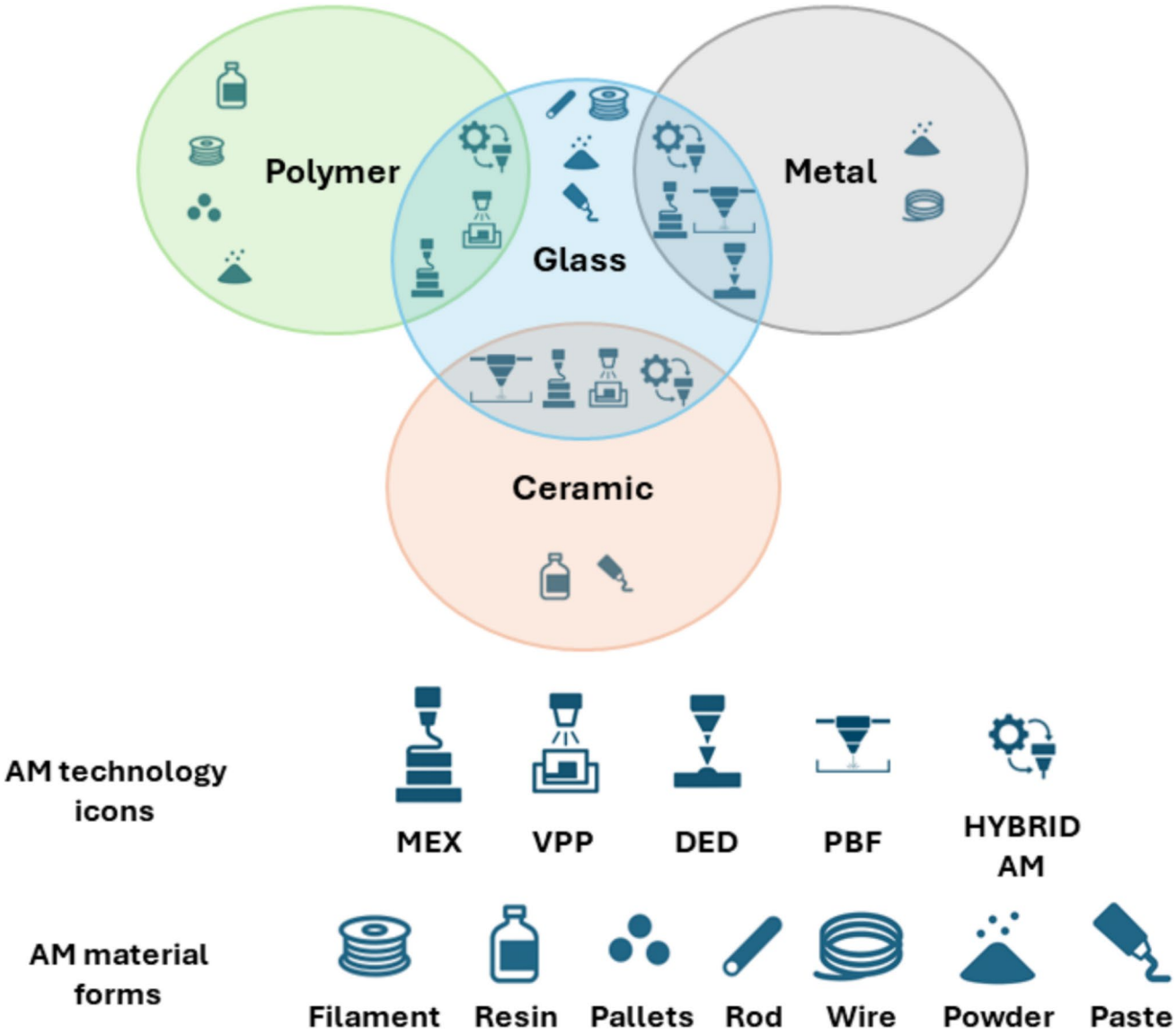


Fig. 5 Conceptual relationship between glass and other AM material domains, illustrating common feedstock forms and possible hybridisation routes

Table 4 Glass materials demonstrated in the literature for different AM categories and technologies

AM category	Technology	Glass materials in the literature
MEX-TRB	TRB (direct melt / filament-fed)	Silicate-based glass, fused quartz, fused silica glass, borosilicate glass, phosphate glass, soda-lime glass, chalcogenide glass [47, 56, 58, 87–90]
MEX-CRB	CRB–DIW	Silica glass, soda-lime glass, bioactive glass [43, 44, 54]
	CRB–paste extrusion	Silica glass, soda-lime glass [55, 91]
VPP	DLP	Fused silica glass, boro-phospho-silicate / multicomponent glass [38, 62, 64, 92]
	SLA	Fused silica glass, boro-aluminosilicate glass, luminescent glass [5, 39, 59, 93]
	TPP	Fused silica / glass-ceramic [17, 45, 61]
DED-LB	DED-W	Fused quartz, fused silica glass, borosilicate glass, soda-lime glass [50, 68, 71, 94]
	DED-P	Soda-lime–silica glass (powder-fed) [74]
PBF-LB	PBF-LB/M (SLM-style)	Soda-lime glass [75–77, 95]
	PBF-LB/P (SLS / selective densification)	Fused silica (selective densification), A–W glass-ceramic (indirect SLS), LAS glass-ceramic (laser densification) [78, 79]

voxel-level spatial control. Filament-based DED also shows potential for structural glass deposition and compositional grading; however, challenges such as thermal mismatch, surface roughness, and interfacial stress currently limit its broader applicability.

In contrast, powder-based approaches are fundamentally limited in their suitability for MMGAM. These techniques rely on static, pre-formulated powder beds and offer no means for dynamic material switching or fine spatial control during printing. Glass powders also pose challenges such as poor flowability, high melting points, and cracking during sintering or cooling, further restricting their use for integrating dissimilar materials within a single structure.

To further contextualise this landscape, Table 5 provides a qualitative comparison of standard AM categories with respect to their demonstrated viability for single-material glass AM and MMGAM. The table distinguishes between (i) Spatial tuning, involving controlled variation of composition, dopants, or refractive index within a continuous glass phase, and (ii) Co-fabrication, the simultaneous deposition of dissimilar material classes (e.g., glass with polymers, metals, or ceramics) within a single structure. using reported feature size, glass-processing temperature range, and demonstrated multi-material capability as key criteria.

Beyond combining different glass compositions, several AM routes also enable hybridisation of glass with polymers, metals, or ceramics. DIW is the most versatile in this respect, supporting co-deposition via multi-ink strategies. VPP enables fine-scale spatial tuning of glass chemistry within a common glass-forming matrix (e.g. dopant/porogen patterning), but true co-fabrication with dissimilar material classes remains uncommon due to resin and post-processing constraints. DED and PBF offer pathways to glass–metal and glass–ceramic integration in high-temperature regimes, yet interfacial cracking, porosity, and residual-stress accumulation currently limit reliability.

The choice of MMGAM strategy ultimately depends on application requirements such as resolution, material compatibility, thermal constraints, and the required mode of integration (spatial tuning vs. co-fabrication). Looking ahead, hybrid platforms that combine complementary printing and consolidation steps, together with advanced interface engineering and thermal process control, are expected

Table 5 Qualitative suitability of AM techniques for glass AM and MMGAM

AM category	Glass AM	MMGAM (i): spatial tuning	MMGAM (ii): co-fabrication	Indicative feature size (bead/track/voxel)	Glass processing / conversion temperature range	Remarks (representative glass-AM studies)
MEX-CRB	Excellent	Excellent	Good	$\sim$ 100–300 μm bead width	$\sim$ 800 –1100 °C(debinding + sintering)	Multi-ink DIW/robocasting enables spatial grading; key limits are shrinkage/warpage and residual porosity [2, 7, 96]
MEX-TRB	Good	Good	Low–moderate	$\sim$ 400 –2000 μm track width	$\sim$ 450 –1200 °C melt temperature	Melt extrusion can yield dense parts; co-fabrication is constrained by viscosity windows, thermal gradients, and CTE mismatch [7, 22, 90]
VPP	Excellent	Excellent (one glass matrix)	Fair–good	sub-μm to $\sim$ 50 μm	$\sim$ 220 –1300 °C (conversion + sintering)	Highest resolution in silica-derived routes; spatial tuning via patterned dopants/porogens is common, while true co-fabrication remains uncommon [5, 7, 97]
DED	Fair–good (early-stage)	Moderate	Moderate	$\sim$ 200 –1500 μm(setup-dependent)	$\gtrsim$ 1400 °C local melt-pool (quartz-like)	Large freeform deposition with feed switching; cracking/stress, roughness, and limited transparency still dominate [7, 22, 50, 68, 71]
PBF-LB	Fair (high geometric complexity; poor transparency)	Low–moderate (glass/glass-ceramic; limited evidence)	Low–moderate	$\sim$ 100–700 μm strut / melt-track scale	$\sim$ 1000 –1400 °C(effective)	Enables highly complex porous/lattice glass architectures and has reported mechanical characterisation; however, current glass PBF-LB typically remains porous/opaque with microcracking and residual-stress limitations, so it is not yet competitive for transparent bulk optics [7, 19, 75, 77, 78, 95]

Table 6 Representative application examples of glass AM processes and MMGAM

AM process	Optics/photonics	Microfluidics/biomedical	Structural/hybrid electronic/other functional components
MEX (CRB / TRB)	Architectural and lighting-grade glass components (e.g. freeform panels, luminaires) [47, 48]	Silica-based microfluidic channels and chip structures [54, 55]	Bioactive and porous glass scaffolds with graded composition or porosity [43, 44]
VPP (SLA / DLP / TPP)	Fused-silica microlenses, waveguides and multicolour photonic architectures via dopant patterning [5, 59]	Fully glass microfluidic chips and 3D microchannel networks [5, 49, 62]	Glass-ceramic microarchitectures and mechanically robust meta-structures obtained by TPP and heat treatment [16, 17, 45]
DED (DED-W / DED-P)	Freeform silica and borosilicate lenses, windows and doped tracks for photonic and sensing functions [68, 71–73]	–	Large structural or shielding components in fused quartz and borosilicate [50, 69, 98]
PBF-LB (SLS / PBF-LB)	Currently focused on opaque structural parts; transparent optical components remain challenging due to porosity and crystallisation [75]	Apatite-wollastonite glass-ceramic bone scaffolds [79]	LAS glass-ceramic substrates and LAS/metal composite components with integrated conductive paths [19, 77, 78]
Hybrid AM	Hybrid AM of transparent silica glass structures with embedded or surface-functionalised regions for optical and photonic use [41, 52]	Lab-on-glass microfluidic systems and hollow fused-silica microchannels combining imprinting, DIW, sacrificial templating and VPP or laser fusion [49, 52]	Alumina-glass composites and photocatalytic glass architectures (e.g. TiO_2-modified scaffolds), as well as glass-metal or glass-ceramic hybrids for electronic and sensor packaging [53, 86]

to play a central role in achieving scalable, defect-tolerant, and functionally integrated glass systems.

4 Applications

Glass AM and, more recently, MMGAM have been explored in four main application areas: (i) optics and photonics, (ii) microfluidics and biomedical devices, (iii) structural and architectural components, and (iv) multifunctional or chemically active systems (e.g. catalytic or bioactive scaffolds). Each AM family contributes differently to this landscape: extrusion-based routes dominate meso-scale devices and scaffolds, VPP leads in high-resolution micro-optical components, DED and PBF target larger structural or glass-ceramic parts, and Hybrid-AM enables more complex multi-material integration. Table 6 summarises representative examples across these domains (for process-specific examples, see Fig. 3).

For MEX, single-material glass AM based on CRB-type processes (DIW and paste extrusion) has mainly been used to fabricate meso-scale lenses, microchannel arrays and bioactive scaffolds [43, 54, 55]. TRB-based melt extrusion platforms such as G3DP extend this to large architectural and lighting components in soda-lime glass, where freeform wall thickness and surface morphology are exploited to combine structural and optical roles [47, 48]. When composition or porosity is deliberately graded within these parts (e.g. bioactive versus non-bioactive regions or refractive-index-tuned segments), these studies also represent early MMGAM-style spatial tuning on the meso scale.

In single-material glass AM, VPP currently provides the most advanced micro-optical and microfluidic demonstrators: SLA/DLP fused-silica nanocomposites enable microlenses, waveguides and fully glass microfluidic chips [5, 49, 62]. On the MMGAM side, spatially resolved doping and multicomponent glass networks allow multicolour luminescent elements and graded-index optics within a common silica backbone [16, 59], making VPP the most mature platform for fine-scale composition tuning in glass.

For DED, glass AM applications to date are dominated by single-material, freeform or large-scale components, such as quartz and borosilicate walls, lenses and shielding elements built from filament-fed CO_2-laser systems [50, 68, 71]. Digital glassforming via volumetric laser heating improves melt homogeneity and supports thicker optical parts or graded compositions within a single build [72].

Powder-bed methods are mainly used for single-material structural and bioactive glass-ceramic applications, including apatite-wollastonite bone scaffolds and LAS substrates [78, 79]. In MMGAM, PBF-LB of LAS/metal composites demonstrates glass-metal co-fabrication with integrated conductive paths [19], although the resulting interfaces are typically opaque and microcracked.

Hybrid-AM broadens the application space by combining glass AM with additional processes or material families. In purely glass-focused hybrid routes, UV-assisted DIW and imprint-extrusion workflows have enabled compact lab-on-glass platforms with embedded microfluidic channels, as well as hollow silica microchannels and transparent components with integrated reservoirs via sacrificial templating and laser-assisted fusion [49, 52, 53, 85]. In genuinely multi-material hybrid-AM, early examples include alumina-glass composites, TiO_2-modified glass scaffolds, and glass-metal or glass-ceramic hybrids for electronic and sensor packaging [86]. Together, these studies illustrate how MMGAM

is beginning to move from single-function optics or scaffolds toward multifunctional devices with coupled optical, mechanical and chemical performance.

Glass AM has begun to appear in commercial products and services, particularly via glass-conversion routes: Glassomer offers fused-silica microfluidic glass chips for diagnostics and chemical analysis [99]. PROTIQ provides 3D-printed glass parts based on sintered Glassomer feedstocks (fused silica) [100]. At the microscale, Nanoscribe commercialised GP-Silica for 2PP fabrication of silica-glass microstructures targeting micro-optics and microfluidics [101].

Overall, current application studies show that MMGAM can deliver functionalities that are difficult or impossible to achieve with conventional glass processing, particularly where spatially varying composition or multi-material integration is required. However, most demonstrations remain at laboratory scale, and systematic studies on long-term performance, upscaling, and standardisation are still largely missing.

5 Challenges and future directions

While multiple AM techniques have demonstrated feasibility for MMGAM, significant hurdles remain, particularly in material compatibility, interfacial bonding, and process control. This section outlines key technical barriers and emerging research directions to guide future advancements in the field.

5.1 Key barriers

MMGAM inherits the difficulties of both glass processing and multi-material AM, so most reported failures can be traced back to a few recurring bottlenecks: thermal compatibility, defect formation, interfacial bonding, and limitations in current machine and process control capabilities [7, 21, 22]. Rather than treating these as isolated issues, it is useful to view them as coupled constraints that jointly define the design space for multi-material glass systems.

High processing temperatures and thermal mismatch Most glass compositions require softening or working temperatures exceeding 1000 °C, and even lower-T_g (glass transition temperature) phosphate or germanate glasses operate far above typical polymer or many metal AM regimes [5, 7]. When glasses with different viscosity-temperature or CTE profiles, or glass and dissimilar materials, are processed in a single build, large thermal gradients and incompatible strain evolution can cause warping, delamination, and microcracking during deposition and cooling [7, 22]. These issues are often magnified in MMGAM because multi-material interfaces experience heterogeneous thermal histories and repeated reheating within complex 3D geometries, increasing residual-stress accumulation and elevating the need for careful thermal management and interface engineering (e.g., preheating, controlled cooling, and compliant or graded interlayers) [7, 80].

Bubble formation and pore defects Bubbles and pores degrade both optical performance and mechanical reliability, and they originate from different mechanisms in different process families. In melt-based routes, bubbles arise from reboiling of volatile species and gas entrapment in the melt pool, whereas in paste- and resin-based systems they are frequently inherited from incomplete debinding or differential shrinkage during sintering [7, 102]. Process variables such as laser energy density, scan speed, ambient pressure, and heat-treatment profiles have been shown to strongly influence bubble density and distribution [22, 72, 102], making robust control particularly challenging in multi-material builds.

Interfacial bonding and compatibility Joining dissimilar glasses or combining glass with polymers, metals, or ceramics is complicated by mismatched surface energies and wetting behaviour, different densification and shrinkage kinetics, and strongly dissimilar thermal profiles. Even in controlled glass-frit bonding, avoiding voids and delamination demands careful matching of CTE, viscosity, and sintering conditions [103]. In MMGAM, typical failure modes include debonding at glass-polymer interfaces due to poor wetting or chemical incompatibility, microcracking at glass-glass boundaries with strong CTE contrast, and interfacial porosity or lack-of-fusion along raster directions where pastes or melts do not fully coalesce [21, 22]. Glass-metal and glass-ceramic joints add further constraints, including oxide-layer and reactivity effects, residual-stress build-up, and the need to accommodate sharp stiffness and CTE mismatches that promote cracking or detachment during cooling [21, 22]. Reported mitigation strategies therefore combine process-window matching with interface engineering, such as compositionally compatible glass backbones with locally tuned dopants and compliant interlayers or functionally graded transitions to reduce abrupt property mismatches [5, 16, 19]. Structural approaches such as surface conditioning, texturing, or tailored roughness can further enhance wetting and mechanical interlocking and promote the formation of transitional interfacial zones [80].

Machine requirements Glass AM demands hardware beyond standard AM, stable high-temperature delivery ($\geq$ 1000 °C) with precise thermal gradients and active annealing, inert or dry atmospheres to prevent devitrification, corrosion-resistant nozzles/refractories, tight motion and extrusion synchronisation for optical-grade surfaces, and reliable multi-material switching with CTE-aware thermal

management [51, 73]. Lack of such capabilities, especially chamber heating/annealing control and durable hot-end materials, remains a primary barrier to reproducible, high-quality components.

Material handling In multi-material AM, different feedstock types must be processed, and transitions between materials introduce risks of cross-contamination and material loss [21]. These challenges can be especially pronounced in powder-fed DED, where material changeovers may require cleaning/purging steps that reduce effective material utilisation and process repeatability [66].

Post-processing. For most glass-AM/MMGAM routes, post-processing (e.g., debinding and sintering for DIW/VPP, or annealing for melt-based routes) is essential to achieve densification, stress relief, and optical quality [7, 22]. In MMGAM, these steps are further complicated by differential shrinkage and CTE mismatch between regions, which can amplify distortion or interfacial cracking unless thermal schedules are co-optimised across materials [21, 22].

Process control and environmental sensitivity Effective MMGAM requires precise control of viscosity, temperature, humidity, and atmospheric composition; even minor fluctuations can lead to incomplete fusion, phase separation, crystallisation, or inconsistent densification [7, 22]. High-temperature operation also introduces additional safety and infrastructure requirements (e.g., enclosed operation for some chemistries, fume handling, and heated chambers/annealing control), increasing system complexity and total cost of ownership [7, 22].

Energy and cost perspective In glass AM, cost is often dominated by thermal processing: direct-melt routes require sustained high-temperature operation and subsequent annealing, while conversion-based DIW/VPP shift much of the energy use and lead time to debinding and sintering [7, 22]. As in AM more broadly, this generally favors low-to-medium production volumes and high-value parts, where eliminating tooling and enabling added functionality or geometric complexity can offset higher process costs.

Lack of standards A major challenge in AM is the absence of standardised protocols for materials, processes, and component-level qualification. For MMGAM, shared reporting and validation practices are particularly needed for thermal histories, defect statistics, and interface performance to improve reproducibility and support industrial adoption.

Overall, MMGAM faces an intricate combination of material, process, and system-level challenges. Among these, thermal compatibility, interfacial bonding, and process stability remain the most pressing. Overcoming these limitations will require coordinated advances in glass formulation, hybrid process design, and real-time monitoring. Future efforts should also focus on predictive simulation frameworks and integrated hybrid-AM systems to enable scalable, defect-tolerant, and functionally optimised glass architectures.

5.2 Future research directions

Future progress in MMGAM will depend on closing the gap between proof-of-concept demonstrations and reproducible, interface-reliable, application-driven manufacturing. The most impactful research directions can be grouped into material design, platform development, interface science, monitoring, and standardisation.

Programmable glass systems Nanoparticle-based inks and resins (MEX-CRB and VPP) already enable spatial tuning of refractive index, luminescence, and porosity by varying dopant content or solids loading within a common backbone [15, 16, 38, 39]. A key next step is to move from one-off formulations toward material sets of co-printable glasses with intentionally aligned (or graded) viscosity-temperature behaviour, CTE, and sintering shrinkage. Such co-processing-compatible material sets would enable sharper functional contrast without sacrificing interfacial integrity, supporting graded-index optics, catalytic/bioactive gradients, and chemically patterned microdevices [18, 22, 43, 93].

Hybrid and multi-stage manufacturing platforms MMGAM will likely advance fastest via platforms that combine additive shaping with targeted consolidation and finishing. Promising routes include staged thermal conditioning and integrated annealing, as well as localised reflow or laser-assisted fusion to heal defects and strengthen interfaces without exposing the full part to extreme thermal cycles [17, 49, 52]. Hybrid workflows that couple AM with replication/forming (e.g., imprint-extrusion and related approaches) are particularly attractive for improving surface quality and dimensional fidelity in glass microdevices [53, 85]. Layer-free volumetric approaches broaden the design space further and may become relevant for multi-material structuring once resin switching, inhibition control, and multi-resin compatibility mature [63].

Interface engineering strategies Interfaces remain the defining bottleneck in MMGAM. Research should shift from qualitative success criteria to measurable interfacial performance. Key priorities include (i) design rules and process maps that connect CTE mismatch, viscosity compatibility, wetting, and shrinkage/strain evolution to cracking and delamination risk; (ii) engineered transitions using graded zones, frit/reactive interlayers, or textured substrates/coatings to promote wetting and mechanical interlocking [19, 21, 22, 103]; and (iii) standardised characterisation spanning microstructure/defect statistics and fracture/adhesion testing on representative geometries. Where feasible, maintaining a common glass-forming backbone (e.g., silica-rich

systems) while locally tuning dopants remains a robust strategy for continuous optical interfaces with functional contrast [5, 16, 59].

In-situ process monitoring Because defects (bubbles, incomplete fusion, phase separation, crystallisation) are highly process-sensitive, in-situ monitoring should evolve toward feedback control. Candidate tools include pyrometry and thermal imaging for melt-based routes, optical coherence tomography for near-surface defect tracking in conversion routes, and spectroscopic methods (e.g., Raman/fluorescence) for local structure or dopant-state verification [102]. The long-term objective is closed-loop control of temperature history and consolidation state at and near interfaces, where reliability is most sensitive.

Expanded glass chemistry MMGAM will benefit from printable formulations beyond silica-rich systems, including bioactive glasses, rare-earth-doped glasses, and glass-ceramic precursors tailored for controlled crystallisation without interfacial embrittlement [17, 43, 89, 91]. Equally important is linking chemistry to manufacturability: composition-dependent sintering, volatilisation, and crystallisation should be treated as co-design constraints alongside optical/electrical functionality.

Advanced application domains The most convincing next demonstrations are those where multi-material integration is essential and performance can be benchmarked: graded-index or multi-wavelength optics; microfluidic-photonic systems with embedded sensing/filtration; catalytic or photocatalytic architectures for harsh environments; and hybrid glass-ceramic/metal platforms for packaging and interconnects [93]. Demonstrators should explicitly report long-term stability (thermal cycling, chemical exposure, fatigue) and interface failure modes.

Standardisation and reporting To accelerate reproducibility and industrial uptake, the community needs shared reporting practices for feedstock composition, thermal history (debinding/sintering/annealing), porosity/bubble quantification, and interface test metrics, together with benchmark artefacts for cross-platform comparison [21, 22].

Sustainability Sustainability studies should quantify energy, yield, and waste trade-offs between melt-based and conversion-based MMGAM routes, and identify pathways for lower-temperature processing, higher reuse/recycling of feedstocks, and safer handling.

Taken together, these directions emphasise a shift from isolated demonstrations toward co-designed material-process-interface systems, supported by quantitative reliability testing and monitoring-enabled process control.

6 Conclusions

Over the past decade, AM has evolved into a transformative production paradigm, offering advantages such as design freedom, material efficiency, and the fabrication of intricate geometries unattainable through conventional manufacturing. Yet, the adoption of AM for glass materials remains constrained by technical challenges related to high processing temperatures, limited material compatibility, and precision requirements.

Recent advancements, particularly in multi-material integration, temperature-controlled deposition, and toolpath optimisation, are beginning to address these barriers. The future of glass AM, and especially MMGAM, holds strong potential in domains such as optics, photonics, microfluidics, and biomedical engineering, where functional integration across material interfaces is critical.

This review has provided a comprehensive overview of the current state of glass AM, including a detailed evaluation of techniques, materials, and application areas. A key focus was placed on MMGAM, addressing the formulation challenges, interfacial compatibility, and system-level design considerations unique to multi-material workflows.

By systematically organizing the available literature, identifying knowledge gaps, and proposing future directions, this work serves as a foundation for advancing both the scientific understanding and practical implementation of MMGAM. Continued research in tailored glass compositions, hybrid AM platforms, and real-time process control will be essential for transitioning these technologies from the lab to industrial-scale applications.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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References

1. Palivela BC, Bandari SD, Mamilla RS (2022) Extrusion-based 3D printing of bioactive glass scaffolds-process parameters and mechanical properties: a review. Bioprinting 27:e00219. https://doi.org/10.1016/j.bprint.2022.e00219
2. Zhang D, Liu X, Qiu J (2021) 3D printing of glass by additive manufacturing techniques: a review. Front Optoelectron 14:263–277. https://doi.org/10.1007/s12200-020-1009-z
3. Zhu Y, Tang T, Zhao S, Joralmon D, Poit Z, Ahire B, Keshav S, Raje AR, Blair J, Zhang Z, Li X (2022) Recent advancements and applications in 3D printing of functional optics. Addit Manuf 52:102682. https://doi.org/10.1016/j.addma.2022.102682
4. Hamed H, Eldiasty M, Seyedi-Sahebari S-M, Abou-Ziki JD (2023) Applications, materials, and fabrication of micro glass parts and devices: an overview. Mater Today 66:194–220. https://doi.org/10.1016/j.mattod.2023.03.005
5. Kotz F, Arnold K, Bauer W, Schild D, Keller N, Sachsenheimer K, Nargang TM, Richter C, Helmer D, Rapp BE (2017) Three-dimensional printing of transparent fused silica glass. Nature 5440(7650):337–339. https://doi.org/10.1038/nature22061
6. Yang Y, Song X, Li X, Chen Z, Zhou C, Zhou Q, Chen Y (2018) Recent progress in biomimetic additive manufacturing technology: from materials to functional structures. Adv Mater 300(36):1706539. https://doi.org/10.1002/adma.201706539
7. Xin C, Li Z, Hao L, Li Y (2023) A comprehensive review on additive manufacturing of glass: recent progress and future outlook. Mater Des 227:111736. https://doi.org/10.1016/j.matdes.2023.111736
8. Leung Y-S, Kwok T-H, Li X, Yang Y, Wang CCL, Chen Y (2019) Challenges and status on design and computation for emerging additive manufacturing technologies. J Comput Inf Sci Eng 190(2):021013. https://doi.org/10.1115/1.4041913
9. Jin L, Zhai X, Wang K, Zhang K, Wu D, Nazir A, Jiang J, Liao W-H (2024) Big data, machine learning, and digital twin assisted additive manufacturing: a review. Mater Des 244:113086. https://doi.org/10.1016/j.matdes.2024.113086
10. Massimino D, Townsend E, Folinus C, Stern M, Becker K (2024) Additive manufacturing of interlocking glass masonry units. Glass Struct Eng 90(3):397–417. https://doi.org/10.1007/s40940-024-00279-8
11. Aliyev R (2025) Planung von additiven Fertigungsprozessen, 1st edn. Springer Vieweg Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-70199-7
12. Gołębiewski P, Wiencław P, Cimek J, Socha P, Pysz D, Filipkowski A, Stepniewski G, Czerwińska O, Kujawa I, Stepień R, Kasztelanic R, Burgs A, Buczyński R (2024) 3D soft glass printing of preforms for microstructured optical fibers. Addit Manuf 79:103899. https://doi.org/10.1016/j.addma.2023.103899
13. Kluck S, Prediger R, Hambitzer L, Nekoonam N, Dreher F, Luitz M, Lunzer M, Worgull M, Schneider M, Rapp BE, Kotz-Helmer F (2024) Sub-micron replication of fused silica glass and amorphous metals for tool-based manufacturing. Adv Sci 110(35):2405320. https://doi.org/10.1002/advs.202405320
14. Rajeshirke M, Fidan I, Naikwadi V, Alkunte S, Gupta A, Mohammadizadeh M (2025) Material extrusion-based multi-material 3d printing: a holistic review of recent advances. Int J Adv Manuf Tech. https://doi.org/10.1007/s00170-025-15920-1
15. Dudukovic NA, Wong LL, Nguyen DT, Destino JF, Yee TD, Ryerson FJ, Suratwala T, Duoss EB, Dylla-Spears R (2018) Predicting nanoparticle suspension viscoelasticity for multimaterial 3D printing of silica-titania glass. ACS Appl Nano Mater 10(8):4038–4044. https://doi.org/10.1021/acsanm.8b00821
16. Barbera L, Madrid-Wolff J, Emma R, Masania K, Boniface A, Loterie D, Delrot P, Moser C, Studart AR (2024) Multimaterial volumetric printing of silica-based glasses. Adv Mater Tech 90(7):2202117. https://doi.org/10.1002/admt.202202117
17. Gailevičius D, Padolskytė V, Mikoliūnaitė L, Šakirzanovas S, Juodkazis S, Malinauskas M (2019) Additive-manufacturing of 3D glass-ceramics down to nanoscale resolution. Nanoscale Horiz 40(3):647–651. https://doi.org/10.1039/c8nh00293b
18. Yang L, Mayer F, Bunz UHF, Blasco E, Wegener M (2021) Multi-material multi-photon 3D laser micro-and nanoprinting. Light Adv Manuf 20(3):296–312. https://doi.org/10.37188/lam.2021.017
19. Cheng D, Wei C, Huang Y, Zhang Z, Wang D, Liu Z, Newman M, Ma T, Chueh Y-H, Zhang X, Liu Z, Li L (2022) Additive manufacturing of lithium aluminosilicate glass-ceramic/metal 3D electronic components via multiple material laser powder bed fusion. Addit Manuf 49:102481. https://doi.org/10.1016/j.addma.2021.102481
20. Hisham M, Salih AE, Butt H (2023) 3D printing of multimaterial contact lenses. ACS Biomater Sci Eng 90(7):4381–4391. https://doi.org/10.1021/acsbiomaterials.3c00175
21. Hasanov S, Alkunte S, Rajeshirke M, Gupta A, Huseynov O, Fidan I, Alifui-Segbaya F, Rennie A (2022) Review on additive manufacturing of multi-material parts: progress and challenges. J Manuf Mater Process 60(1):4. https://doi.org/10.3390/jmmp6010004
22. Shaw M, Vlahopoulos D, Usma-Mansfield C, Sarmiento J, Snooks R, Koutsonikolas T, Birbilis N (2025) The additive manufacturing of glass a critical review. Appl Sci 150(6):1324. https://doi.org/10.3390/app15063414
23. Scheithauer U, Johne R, Gottlieb L, Weingarten S, Wiemer H, Holtzhausen S (2025) Recent developments in the field of additive manufacturing of ceramic-based multi-material components open the door to highly functionalized components. Prog Addit Manuf. https://doi.org/10.1007/s40964-025-01112-6
24. Chen Y, Ye P, Huang L, Zhao S, Zhang H, Wang Z, Liu Y, Liu H (2024) Photopolymerized 3d printing materials for optical elements. Adv Opt Mater 120(12):2302685. https://doi.org/10.1002/adom.202302685
25. Arriaga-Dávila JA, Li X, Guerra E, Wegener M, Barbera L (2025) From single to multi-material 3d printing of glass-ceramics for micro-optics. Small Methods. https://doi.org/10.1002/smtd.202301626
26. Lakhdar Y, Tuck C, Binner J, Terry A, Goodridge R (2021) Additive manufacturing of advanced ceramic materials. Prog Mater Sci 116:100736. https://doi.org/10.1016/j.pmatsci.2020.100736
27. Schott AG (2025) Technical data for ar-glas® soda-lime glass tubing. URL https://www.schott.com/en-gb/products/ar-glas-p1000366/technical-details. Accessed 12 Dec 2025
28. Corning Inc. 1737f alkali-free boro-aluminosilicate glass technical sheet (2024). https://www.pgo-online.com/intl/1737.html. Accessed 12 Dec 2025
29. Hilgenberg GmbH (2022) Technical datasheet – glass 0100 (soda-lime). https://www.hilgenberg-gmbh.de. Version 05/2022. Accessed 12 Dec 2025
30. Schott (2017) Duran® borosilicate glass tubing. https://www.schott.com/en-gb/products/duran-p1000368/downloads. Accessed 12 Dec 2025
31. Präzisions Glas, Optik GmbH. Soda-lime float glass with sight glass quality, (2025). https://www.pgo-online.com/intl/soda-lime-glass.html. Accessed 25 July 2025
32. Corning Inc. Hpfs® product brochure – fused silica grades 7979, 7980, and 8655 (2015). https://www.corning.com/media/worldw

ide/csm/documents/HPFS_Product_Brochure_All_Grades_2015_07_21.pdf. Accessed 12 Dec 2025
33. Mauro JC, Yue Y, Ellison AJ, Gupta PK, Allan DC (2009) Viscosity of glass forming liquids. Proc Natl Acad Sci 1060(47):19780–19784. https://doi.org/10.1073/pnas.0911705106
34. Doremus RH (1991) Glass: an overview. Concise Encyclopedia of Advanced Ceramic Materials, pages 166–170. https://doi.org/10.1016/B978-0-08-034720-2.50052-6
35. Corning Inc. Corning 9754 optical glass: Product information and properties (2023). https://www.corning.com/optical-materials. Accessed 12 Dec 2025
36. Zhang K, Bai J, Wan X, Dong L, Wang S, Pan H, Chu Y, Ren J, Zhang J, Peng G-D (2024) Facile fabrication of silica glass embedded with NiO nanoparticles by 3D printing technology and its optical nonlinearity. Adv Photonics Res 50(2):2300244. https://doi.org/10.1002/adpr.202300244
37. Arita R, Iijima M, Fujishiro Y, Morita S, Furukawa T, Tatami J, Maruo S (2020) Rapid three-dimensional structuring of transparent SiO_2 glass using interparticle photo-cross-linkable suspensions. Commun Mater 10(1):30. https://doi.org/10.1038/s43246-020-0029-y
38. Moore DG, Barbera L, Masania K, Studart AR (2020) Three-dimensional printing of multicomponent glasses using phase-separating resins. Nat Mater 190(2):212–217. https://doi.org/10.1038/s41563-019-0525-y
39. Ouyang M, Zhang H, Li M, Zhang J, Gong Y, Huang X, Liu X, Qiu J, Yang Z, Dong G (2023) 3D printing of luminescent glass with controlled distribution of emission colors for multi-dimensional optical anti-counterfeiting. Laser Photonics Rev 170(8):2300068. https://doi.org/10.1002/lpor.202300068
40. Polsakiewicz D, Kollenberg W (2015) Comparison of silver sources for silver/glass compounds by multi-material 3D-printing. Additional Conferences (Device Packaging, HiTEC, HiTEN, and CICMT), 20150 (CICMT): 000305–000313. https://doi.org/10.4071/CICMT-THA31
41. Mingareev I, Bautista N, Yadav D, Bhanderi A (2023) Ultrafast laser deposition of powder materials on glass. Opt Laser Tech 160:109078. https://doi.org/10.1016/j.optlastec.2022.109078
42. Roitero E, Boterff JL, Chevalier J, Marquette C, Chenal J-M, Coutial E-J (2025) Addressing challenges in additive manufacturing of polylactic acid-p (l, dl) la-and bio-glass composites for implantable medical devices. Prog Addit Manuf. https://doi.org/10.1007/s40964-025-01059-8
43. Ben-Arfa BAE, Neto AS, Palamá IE, Salvado IMM, Pullar RC, Ferreira JMF (2019) Robocasting of ceramic glass scaffolds: sol-gel glass, new horizons. J Eur Ceram Soc 390(4):1625–1634. https://doi.org/10.1016/j.jeurceramsoc.2018.11.019
44. Duncan B, Beck D, Miller P, Benz R, Ledford K, Rosh-Gorsky A, Smith M (2024) Low-temperature additive manufacturing of silica glass composites. ACS Mater Lett 6:720–728. https://doi.org/10.1021/acsmaterialslett.3c01510
45. Doualle T, André J-C, Gallais L (2021) 3D printing of silica glass through a multiphoton polymerization process. Opt Lett 460(2):364–367. https://doi.org/10.1364/OL.414848
46. Additive manufacturing: general principles—fundamentals and vocabulary (2021). https://www.iso.org/standard/74514.html. Vocabulary and taxonomy for AM
47. Klein J, Stern M, Franchin G, Kayser M, Inamura C, Dave S, Weaver JC, Houk P, Colombo P, Yang M, Oxman N (2015) Additive manufacturing of optically transparent glass. 3D Print Addit Manuf 20(3):92–105. https://doi.org/10.1089/3dp.2015.0021
48. Wang Y, Gawedzinski J, Pawlowski ME, Tkaczyk TS (2018) 3D printed fiber optic faceplates by custom controlled fused deposition modeling. Opt Express 260(12):15362–15376. https://doi.org/10.1364/OE.26.015362
49. Kotz F, Risch P, Arnold K, Sevim S, Puigmartí-Luis J, Quick A, Thiel M, Hrynevich A, Dalton PD, Helmer D, Rapp BE (2019) Fabrication of arbitrary three-dimensional suspended hollow microstructures in transparent fused silica glass. Nat Commun 100(1):1439. https://doi.org/10.1038/s41467-019-09497-z
50. Sleiman K, Sleiman O, Rettschlag K, Jäschke P, Kaierle S (2024) Experimental investigations on the formation of boundary layers in glass-based additive manufacturing of fused silica fibers by laser glass deposition. Int J Adv Manuf Tech. https://doi.org/10.1007/s00170-024-14586-5
51. Peters D, Drallmeier J, Bristow DA, Landers RG, Kinzel E (2018) Sensing and control in glass additive manufacturing. Mechatronics 56:188–197. https://doi.org/10.1016/j.mechatronics.2018.06.002
52. Zhang H, Zhang Q, Xiang P, Wang Y, Lei J (2024) Direct additive manufacturing of transparent fused silica glasses using an integrated paste printing and laser in-situ melting technology. Addit Manuf 84:104135. https://doi.org/10.1016/j.addma.2024.104135
53. Marzi AD, Giometti G, Erler J, Colombo P, Franchin G (2022) Hybrid additive manufacturing for the fabrication of freeform transparent silica glass components. Addit Manuf 54:102727. https://doi.org/10.1016/j.addma.2022.102727
54. Nguyen DT, Meyers C, Yee TD, Dudukovic NA, Destino JF, Zhu C, Duoss EB, Baumann TF, Suratwala T, Smay JE, Dylla-Spears R (2017) 3D-printed transparent glass. Adv Mater 290(26):1701181. https://doi.org/10.1002/adma.201701181
55. Derevianko OV, Derevianko OV, Zakiev VI, Zgalat-Lozynskyy OB (2022) 3D printing of porous glass products using the robocasting technique. Powder Metall Met Ceram 600(9):546–555. https://doi.org/10.1007/s11106-022-00267-z
56. Mader M, Hambitzer L, Schlautmann P, Jenne S, Greiner C, Hirth F, Helmer D, Kotz-Helmer F, Rapp BE (2021) Melt-extrusion-based additive manufacturing of transparent fused silica glass. Adv Sci 80(23):2103180. https://doi.org/10.1002/advs.202103180
57. Utz J, Ferbert A, Geis N, Ruckdäschel H (2025) Mechanical properties and predictive analysis of multi-material polypropylene-glass fiber sandwich structures produced by material extrusion. Polym Compos. https://doi.org/10.1002/pc.29617
58. Carcreff J, Cheviré F, Galdo E, Lebullenger R, Gautier A, Adam JL, Coq DL, Brilland L, Chahal R, Renversez G, Troles J (2021) Mid-infrared hollow core fiber drawn from a 3D printed chalcogenide glass preform. Opt Mater Express 110(1):198–209. https://doi.org/10.1364/OME.415090
59. Liu C, Qian B, Ni R, Liu X, Qiu J (2018) 3D printing of multicolor luminescent glass. RSC Adv 80(55):31564–31567. https://doi.org/10.1039/C8RA06706F
60. Zhang D, Xiao W, Liu C, Liu X, Ren J, Xu B, Qiu J (2020) Highly efficient phosphor-glass composites by pressureless sintering. Nat Commun 110(1):2805. https://doi.org/10.1038/s41467-020-16649-z
61. Wen X, Zhang B, Wang W, Ye F, Yue S, Guo H, Gao G, Zhao Y, Fang Q, Nguyen C, Zhang X, Bao J, Robinson JT, Ajayan J, Pulickel M, Lou J (2021) 3D-printed silica with nanoscale resolution. Nat Mater 200(11):1506–1511. https://doi.org/10.1038/s41563-021-01111-2
62. Toombs JT, Luitz M, Cook CC, Jenne S, Li CC, Rapp BE, Kotz-Helmer F, Taylor HK (2022) Volumetric additive manufacturing of silica glass with microscale computed axial lithography. Science 3760(6590):308–312. https://doi.org/10.1126/science.abm6459
63. Thompson JE (2025) Volumetric additive manufacturing: ushering in a new era of fabrication. Hardware 3(1):2. https://doi.org/10.3390/hardware3010002
64. Hatscher T, Uta L , Yazici A-O, Risch P, Kotz-Helmer F, Frosz MH (2024) Fused silica microstructured optical fibers made from

3D printed nanocomposite resin. Laser 3D Manuf XI 12876:138–142. SPIE. https://doi.org/10.1117/12.3001669
65. Shah A, Gupta NK, Aliyev R, Zeidler H (2024) Study of various process parameters on bead penetration and porosity in wire arc additive manufacturing (waam) of copper alloy cu1897. Appl Sci 140(20):9188. https://doi.org/10.3390/app14209188
66. Ahn D-G (2021) Directed energy deposition (ded) process: state of the art. Int J Precis Eng Manuf-Green Tech 80(2):703–742. https://doi.org/10.1007/s40684-020-00302-7
67. Von Witzendorff P, Pohl L, Suttmann O, Heinrich P, Heinrich A, Zander J, Bragard H, Kaierle S (2018) Additive manufacturing of glass: Co2-laser glass deposition printing. Proced CIRP 74:272–275. https://doi.org/10.1016/j.procir.2018.08.109
68. Luo J, Hostetler JM, Gilbert L, Goldstein JT, Urbas AM, Bristow DA, Landers RG, Kinzel EC (2018) Additive manufacturing of transparent fused quartz. Opt Eng 570(4):041408–041408. https://doi.org/10.1117/1.OE.57.4.041408
69. Capps NE, Goldstein JT, Rettschlag K, Sleiman K, Jaeschke P, Kaierle S, Kinzel EC (2022) Direct laser heating of the filament/substrate interface in digital glass forming. Manuf Lett 31:106–109. https://doi.org/10.1016/j.mfglet.2021.08.005
70. Hong Z, Luo T, Jiang S, Liang R (2023) Fiber-fed 3D printing of germanate glass optics. Photonics 10(4):378. https://doi.org/10.3390/photonics10040378
71. Fröhlich F, Hildebrand J, Bergmann JP (2022) Additive manufacturing with borosilicate glass and soda-lime glass. In 2nd Int Conf Adv Joining Processes (AJP 2021) 151–164. Springer. https://doi.org/10.1007/978-3-030-95463-5_10
72. Deutsch-Garcia L, Salinas-Sáenz S, Hlifka B, Ahuett-Garza H, Anders R, Kinzel E (2024) Volumetric heating in digital glass forming. J Manuf Process 122:112–119. https://doi.org/10.1016/j.jmapro.2024.05.043
73. Capps N, Zhu C, Goldstein J, Kinzel E (2023) Digital glass forming of photonics. Opt Eng 620(7):076101–076101. https://doi.org/10.1117/1.OE.62.7.076101
74. Spirrett F, Datsiou KC, Magallanes M, Ashcroft I, Goodridge R (2023) Powder-fed directed energy deposition of soda lime silica glass on glass substrates. J Am Ceram Soc 1060(1):227–240. https://doi.org/10.1111/jace.18638
75. Luo J, Pan H, Kinzel EC (2014) Additive manufacturing of glass. J Manuf Sci Eng 1360(6):061024. https://doi.org/10.1115/1.4028531
76. Fateri M, Gebhardt A, Thuemmler S, Thurn L (2014) Experimental investigation on selective laser melting of glass. Phys Proced 56:357–364. https://doi.org/10.1016/j.phpro.2014.08.118
77. Datsiou KC, Spirrett F, Ashcroft I, Magallanes M, Christie S, Goodridge R (2021) Laser powder bed fusion of soda lime silica glass: Optimisation of processing parameters and evaluation of part properties. Addit Manuf 39:101880. https://doi.org/10.1016/j.addma.2021.101880
78. Zocca A, Colombo P, Günster J, Mühler T, Heinrich JG (2013) Selective laser densification of lithium aluminosilicate glass ceramic tapes. Appl Surf Sci 265:610–614. https://doi.org/10.1016/j.apsusc.2012.11.058
79. Xiao K, Dalgarno KW, Wood DJ, Goodridge RD, Ohtsuki C (2008) Indirect selective laser sintering of apatite-wollostonite glass-ceramic. Proc Inst Mech Eng Part H J Eng Med 2220(7):1107–1114. https://doi.org/10.1243/09544119JEIM411
80. Yang T, Qiu Y, Han W, Kong L (2025) Investigation of laser powder bed fusion for optically non-uniform quartz glass. Prog Addit Manuf. https://doi.org/10.1007/s40964-025-01049-w
81. Karunakaran KP, Gupta NK, Patel AK, Rakeshkumar K, Ganesan G, Sealy SM, Bernard A (2022) Multi-station multi-axis hybrid layered manufacturing (msma-hlm). Manuf Lett 33:630–639. https://doi.org/10.1016/j.mfglet.2022.07.078
82. Oropeza D, Firdosy S, Hofmann DC (2022) Development of in-plane ss316 to m300 maraging steel gradients via directed energy deposition. Addit Manuf Lett 3:100078. https://doi.org/10.1016/j.addlet.2022.100078
83. Schneck M, Horn M, Schmitt M, Seidel C, Schlick G, Reinhart G (2021) Review on additive hybrid-and multi-material-manufacturing of metals by powder bed fusion: state of technology and development potential. Prog Addit Manuf 60(4):881–894. https://doi.org/10.1007/s40964-021-00205-2
84. Gupta NK, Ganesan G, Siddhartha S, Karade SR, Paul AK, Dubey S, Ely RH, Karunakaran KP (2025) In situ pre-heating in wire arc additive manufacturing: Design, development, and experimental investigation on residual stresses and metallurgical and mechanical properties. J Mater Eng Perform 340(12):12256–12267. https://doi.org/10.1007/s11665-024-10011-w
85. Haq MR, Mazinani B, Subramanian V, Schift H (2024) Room temperature imprinting of water-based microparticulate inks for realizing glass microfluidic channels. Mater Des 242:112982. https://doi.org/10.1016/j.matdes.2024.112982
86. Zhang W, Melcher R, Travitzky N, Bordia RK, Greil P (2009) Three-dimensional printing of complex-shaped alumina/glass composites. Adv Eng Mater 110(12):1039–1043. https://doi.org/10.1002/adem.200900213
87. Brun P-T, Inamura C, Lizardo D, Franchin G, Stern M, Houk P, Oxman N (2017) The molten glass sewing machine. Philos Trans Royal Soc A Math Phys Eng Sci 3750(2093):20160156. https://doi.org/10.1098/rsta.2016.0156
88. Inamura C, Stern M, Lizardo D, Houk P, Oxman N (2018) High fidelity additive manufacturing of transparent glass structures. In: Proc IASS Annual Symposia. vol 2018, International Association for Shell and Spatial Structures (IASS), 2018, p 1–8
89. Baudet E, Ledemi Y, Larochelle P, Morency S, Messaddeq Y (2019) 3D-printing of arsenic sulfide chalcogenide glasses. Opt Mater Express 90(5):2307–2317. https://doi.org/10.1364/OME.9.002307
90. Zaki RM, Strutynski C, Kaser S, Bernard D, Hauss G, Faessel M, Sabatier J, Canioni L, Messaddeq Y, Danto S, Cardinal T (2020) Direct 3D-printing of phosphate glass by fused deposition modeling. Mater Des 194:108957. https://doi.org/10.1016/j.matdes.2020.108957
91. Huang TS, Rahaman MN, Doiphode ND, Leu MC, Sonny Bal B, Day DE, Liu X (2011) Freeze extrusion fabrication of 13–93 bioactive glass scaffolds for repair and regeneration of load-bearing bones. J Mater Sci Mater Med 22:515–523. https://doi.org/10.1007/s10856-011-4236-4
92. Zhu D, Zhang J, Qiao X, Li Y (2024) 3D printing of glass aspheric lens by digital light processing. J Manuf Process 116:40–47. https://doi.org/10.1016/j.jmapro.2024.02.038
93. Mehta A, Ozog P, Dasan A, Kraxner J, Elsayed H, Grigolato L, Galusek D, Bernardo E (2024) Low cost structured photocatalysts from stereolithography of colorless pharmaceutical glass. J Eur Ceram Soc 440(9):5480–5489. https://doi.org/10.1016/j.jeurceramsoc.2023.12.080
94. Luo J, Gilbert LJ, Qu C, Morrow B, Bristow DA, Landers RG, Goldstein J, Urbas A, Kinzel EC (2015) Solid freeform fabrication of transparent fused quartz using a filament fed process. In: Proc 26th Annual Intl Solid Freeform Fabrication Symp (2015, Austin, TX)
95. Fateri M, Gebhardt A (2015) Selective laser melting of soda-lime glass powder. Int J Appl Ceram Technol 120(1):53–61. https://doi.org/10.1111/ijac.12338
96. Nan B, Gołbiewski P, Buczyński R, Galindo-Rosales FJ, Ferreira JMF (2020) Direct ink writing glass: a preliminary step for optical application. Materials 130(7):2020. https://doi.org/10.3390/ma13071636

97. Li M, Yue L, Rajan AC, Yu L, Sahu H, Montgomery SM, Ramprasad R, Qi HJ (2023) Low-temperature 3d printing of transparent silica glass microstructures. Sci Adv 90(40):eadi2958. https://doi.org/10.1126/sciadv.adi2958
98. Chhadeh PA, Sleiman K, Hoffmann H, Funke N, Rettschlag K, Jäschke P, Baitinger M, Knaack U, Kaierle S, Seel M (2025) Design of additively manufactured glass components for glass point fixings. Glass Struct Eng 100(1):6. https://doi.org/10.1007/s40940-024-00290-z
99. Glassomer. Microfluidics: fused silica glass chips. https://www.glassomer.com/products/microfluidic-chips.html
100. PROTIQ. 3d-printing glass. https://www.protiq.com/en/3d-printing/material/glass/
101. Nanoscribe. Gp-silica: photoresin for 3d printed fused silica glass microstructures. https://www.nanoscribe.com/en/products/2pp-printing-materials/photoresin-gp-silica/
102. Luo J, Gilbert LJ, Peters DC, Bristow DA, Landers RG, Goldstein JT, Urbas AM, Kinzel EC (2016) Bubble formation in additive manufacturing of glass. In Advanced Optics for Defense Applications: UV through LWIR 9822:86–91. SPIE. https://doi.org/10.1117/12.2224321
103. Liu C, Oriekhov T, Fokine M (2022) Investigation of glass bonding and multi-layer deposition during filament-based glass 3d printing. Front Mater 9:978861. https://doi.org/10.3389/fmats.2022.978861
104. Hong, Luo Yingying, Zhang Jianwu, Yu Xinran, Dong Tianfeng, Zhou (2023) Additive subtractive and formative manufacturing of glass-based functional micro/nanostructures: A comprehensive review Materials & Design 233112285-10.1016/j.matdes.2023.112285

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